



Public trust, perceived accuracy, perceived likelihood, and concern on multi-model climate projections communicated with different formats

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Abstract

Scientific uncertainties in climate change projections are generally addressed using an ensemble method, in which multiple models are used to generate climate projections. In the interest of transparent and honesty, such uncertainty should be communicated to the general public. Thus, it is important to investigate how such uncertainty should be communicated to the general public. This study explored three uncertainty representation formats—average, range, and multi-value—to investigate how each format affected the general public's trust, perceived accuracy, perceived likelihood, and concern after acknowledging the presence of uncertainty in climate projections (i.e., the use of multi-model climate projections). We conducted a web survey of 2400 participants in Japan, in which we randomly assigned each participant to one of three formats by which climate projection uncertainty was presented. We then asked participants to rate trust, perceived accuracy, perceived likelihood, and concern regarding the climate projections. The multi-value format enhanced trust and perceived accuracy and partially increased perceived likelihood and concern regarding the climate projections compared to the average and range formats, regardless of participants' numeracy and education level. This study suggests that the multi-value format might be effective for communicating multi-model projections and promoting public trust and support for climate policies.

Keywords Climate change · Climate projection · Uncertainty · Risk communication · Trust · Concern

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1 Introduction

To limit global warming to below 2 °C relative to pre-industrial levels is a policy goal widely accepted by many countries and international organizations. However, achieving this goal requires public support for climate mitigation policies that will reduce greenhouse gas emissions. Information on predicted outcomes of mitigation policies is crucial for the public to support policies. Mitigation policy outcomes are inevitably uncertain, because climate change projections involve the highly complex and nonlinear climate system, as well as insufficient data, which results in scientific uncertainty. Ensemble methods are often used to address scientific uncertainty in climate change projections (IPCC 2013; Tebaldi and Knutti 2007). An ensemble method is a method to predict probabilities using a collection of climate models with various key factors such as initial conditions, parameters, and structures. Each model included in the ensemble produces different projections. Thus, the public faces uncertainty regarding whether to support any given climate mitigation policy due to the different projections of climate models (Fischhoff and Davis 2014; Kunreuther et al. 2013). In the interest of transparency and honesty, such uncertainty should be communicated to the general public.

Generally, a range of projections (typically the best and worst projections) is often used to communicate the uncertainty of climate model projections to general public. We term this the “range” format of uncertainty representation. In health and environmental risk literature, numerous studies have investigated the effect on general public responses of the range format compared to a point estimate when communicating uncertainty (e.g., Han et al. 2009, 2011; Politi et al. 2007; Schapira et al. 2001; Spiegelhalter 2017; Spiegelhalter et al. 2011; Trevena et al. 2013). Although the evidence is mixed, communicating uncertainty using the range format generally leads people to consider communicators (e.g., scientists or governments) to be honest but incompetent (Jensen 2008; Johnson 2003; Johnson and Slovic 1995) and increases their perceived risk (Han et al. 2009; Viscusi 1997) and worry (Han et al. 2009, 2011; Johnson and Slovic 1995) compared to communicating a point estimate. A few studies have examined the effect of the range format on psychological responses. Joslyn and LeClerc (2016) found that communicating uncertainty using the range format increases general public trust and concern regarding climate projections compared to conveying a point estimate. Similarly, Howe et al. (2019) found that use of the range format increased general public trust of climate scientists compared to communicating a point estimate.

These previous studies have focused on differences in public responses to communication that did or did not convey uncertainty. However, when climate projections are obtained from an ensemble method, it would be dishonest to present such projections to the general public without acknowledging uncertainty (i.e., differences in climate projections among models). If it is necessary to communicate uncertainty, the question remains as to the format by which uncertainty should be conveyed. Different formats may engender different psychological responses in the general public, such as trust, perceived accuracy, perceived likelihood, and concern regarding climate projections. It is important to explore the effect of the format by which uncertainty is conveyed because numerous previous studies have found that individuals’ support for climate change policy is influenced by their trust of data, the perceived accuracy and perceived likelihood of projections, and the individual’s level of concern regarding climate projections (e.g., Dietz et al. 2007; Drews and van den Bergh 2015; Mossler et al. 2017; O’Connor et al. 1999, 2002; Smith and Leiserowitz 2013; Spence et al. 2011; van der Linden 2017; Zahran et al. 2006).

Although many studies have examined the range format, few studies have explored other formats by which to communicate uncertainty (e.g., Benjamin and Budescu 2018; Bosetti

et al. 2017; Budescu et al. 2014; Howe et al. 2019; Patt 2007). The current study considered two other formats. The first is to present multi-model projections directly to the public. This will be referred to here as “multi-value” format. To our knowledge, only Bosetti et al. (2017) have examined the effect of conveying multi-model projections of climate change on the risk perceptions of climate policy negotiators. They used a boxplot with points representing all climate models’ estimates to convey multi-model projections and compared it with two other formats: boxplot only and boxplot with outliers. They found that the boxplot with points made negotiators’ subjective probability of the increase in global temperature more consistent with the scientific projection than the other formats did. The second is an “average” format, in which the average of multiple projections is communicated to general public, while acknowledging the uncertainty of climate projections (i.e., the existence of different models that produce different climate projections). Note that the average format differs from a point estimate in that the former explicitly acknowledges the uncertainty of climate projections before presenting the average projection to the general public.

The average, range, and multi-value formats have advantages and disadvantages for communicating multi-model projections to the general public. The average format is easily understood, but it might misguide the public to perceive no uncertainty in policy outcome projections, even though the format acknowledges the uncertainty of climate projections. The range format explicitly conveys scientific uncertainty, but it might lead the public to focus on only the best or worst projection (Spiegelhalter et al. 2011). Furthermore, this format might encourage laypersons to perceive the underlying distribution as uniform (Dieckmann et al. 2015; Rinne and Mazzocco 2013) or as they wish to see it (Dieckmann et al. 2017). Because the average and range formats explicitly detail only one or two among multiple model projections, the public might believe that these formats hide aspects of the underlying data. The multi-value format might represent an improvement, in that it provides the most accurate and transparent information regarding projections. However, this format presents complex and detailed information that might be incomprehensible to some individuals.

The advantages and disadvantages of the different formats may be associated with individuals’ cognitive skills. One of the most relevant cognitive skills is likely numeracy. Numeracy is an individual’s ability to understand and use numbers in decision-making tasks (Peters 2012; Reyna et al. 2009), which is not directly related to education level (Rothman et al. 2008; Reyna et al. 2009). Several studies have shown that people with low numeracy had more varied responses than those with high numeracy for essentially the same contents but in different formats, such as percentage vs. frequency (Dieckmann et al. 2009; Peters et al. 2006, 2011) and relative risk vs. absolute risk (Sheridan and Pignone 2002; Sheridan et al. 2003). Dieckmann et al. (2015, 2017) found that people with higher numeracy were more likely to perceive a normal distribution of projections from the range format, whereas Rinne and Mazzocco (2013) found that more numerate people were more prone to infer a uniform distribution of projections from the range format. Han et al. (2011) found that numeracy might not moderate the effect of communicating uncertainty on risk perception and worry for cancer risks.

Education level might also moderate the effects of formats used to communicate uncertainty on general public responses, because education level is related to cognitive skills (Ceci 1991; Ritchie and Tucker-Drob 2018). However, several previous studies obtained mixed results. Schapira et al. (2001) found that less-educated women who were presented with the range format expressed decreased trust of estimates of breast cancer risk compared to when presented with a point estimate. Joslyn and LeClerc (2016) found that the range format increased trust among highly educated people but

not among less-educated people, compared to communicating a point estimate. Howe et al. (2019) found that education level did not significantly influence the effect of the range format on public trust of climate scientists. Regarding concern for climate change, Joslyn and LeClerc (2016) found that education level did not moderate the effect of format.

In sum, regarding the differences in the general public's psychological responses with and without communicating multi-model projections, previous studies found that the range format presenting the best and worst estimates generally increased risk perception, concern, and trust on climate model projections compared to just presenting the point estimate. There are mixed results regarding whether individuals' numeracy or education level moderates the differences in their psychological responses with and without communicating multi-model projections. However, to our knowledge, few studies have examined the effects of different formats for communicating multi-model projections on individuals' psychological responses after acknowledging the uncertainty of climate projections.

This study aimed to investigate how different formats conveying multi-model projections affect general public's psychological responses (trust, perceived accuracy, perceived likelihood, and concern), given that acknowledging the uncertainty of climate projections is indispensable to be honest and transparent in climate risk communication. Due to the exploratory nature of this research, we focused on the most common format and two extreme ones. The range format has been widely used for communicating multi-model projections in numerous documents including the IPCC reports. The average format is simplest and least informative, whereas a multi-value format is most complex and most informative. Furthermore, we aimed to examine whether their numeracy or education level moderates the effect of each format on trust, perceived accuracy, perceived likelihood, and concern. If numeracy or education level influences the effect of format, then the format should be chosen according to the numeracy and education level of the target population. Accordingly, this study explored the following two research questions. **RQ1:** How do three formats (average, range, and multi-value format) affect the general public's trust, perceived accuracy, perceived likelihood, and concern regarding climate projections? **RQ2:** How do numeracy and education level influence the effect of the three formats on trust, perceived accuracy, perceived likelihood, and concern?

2 Methodology

2.1 Survey

We implemented a web survey from October 16 to 18, 2018, after conducting a web-based pre-test with 300 people and extensive pre-analysis. There were 2400 respondents from a sample of 1.9 million people in a commercial dataset (Cross Marketing, Japan). The marketing company recruited participants for this survey by announcing it to all members as a "survey of your opinion" to avoid self-selection bias (i.e., people with high environmental consciousness might participate in a survey regarding climate change as opposed to those who do not identify themselves thus). Any individual aged 20–69 could participate in this survey. The survey participants received reward points that could be exchanged for money. The sample had a mean age of 44.3 years, mean income of 5.71 million yen per year, was composed of 49% females, and 48% university graduates. According to national surveys in Japan, the values for the population overall are 46.4, 5.46 million, 51%, and

49%, respectively.¹ That is, our sample was similar to the general population in these demographic measures.

Figure 1 illustrates the survey processes. The survey queried participants' sociodemographic characteristics (age, gender, education, and income) and presented five stylized facts² regarding climate change, which were obtained from the IPCC (2014) and Stern's (2007) review. Multi-model projections regarding future climate change after implementing hypothetical climate policies were presented using a randomly chosen format (average, range, or multi-value). Then, the survey assessed psychological responses (trust, perceived accuracy, perceived likelihood, and concern) to the multi-model projections. Participants who were presented with the average or range formats were asked to provide their perception of the projections other than the ones provided. Finally, we measured participants' numeracy.

2.2 Multi-model projections of climate change

This study considered two cases of multi-model projections of climate change after implementing hypothetical passive and aggressive climate mitigation policies. The passive policy maintained the annual global greenhouse gas (GHG) emissions at current levels (49 gigatons in 2010) until 2050. The aggressive policy cut emissions by 50%, and it will maintain them at that level until 2050. The possible outcomes of each policy were represented by 19 different probabilities of global temperature increase exceeding 2 °C relative to pre-industrial levels in 2100. These projections were obtained from Meinshausen et al. (2009), who collected 19 published distributions of equilibrium climate sensitivity and provided 19 probability distributions of the likelihood that the rise in the average global temperature would exceed 2 °C relative to pre-industrial levels by 2100. The [supplementary materials](#) in their paper provide an Excel sheet that enabled us to project 19 probabilities of exceeding 2 °C relative to pre-industrial levels in 2100, given the implementation of passive and aggressive hypothetical policies.

In the survey, participants were randomly divided into three groups, each comprising 800 individuals. Each group was assigned to one of the three formats (average, range, and multi-value) used to communicate the multi-model projections of the probability of exceeding a 2 °C temperature increase in 2100 after implementing either the passive or aggressive policy. For all formats, we explained the scientific uncertainty of the climate model projections to participants using the following text: "According to a paper published by Nature, a worldwide leading scientific journal, the probability that the rise in the global average surface temperature will exceed 2 °C in 2100 can be projected using 19 different climate models," and "These multiple projections exist because the best available science cannot provide a unique, precise projection."

¹ The mean age and the proportion of females in the population were obtained from the 2015 Census (Statistics Bureau, Ministry of International Affairs and Communications, Japan), university graduation ratio was obtained from the 2015 School Basic Survey (Ministry of Education, Culture, Sports, Science and Technology, Japan), and income was obtained from the 2016 Comprehensive Survey of Living Conditions (Ministry of Health, Labour and Welfare, Japan).

² The five facts were (1) the trend of the average global temperature, (2) the trend of the average global sea level, (3) the change in the ice surface area of the North Pole, (4) current negative impacts of global warming, and (5) predicted damage if the global mean temperature increase exceeds 2 °C. To ensure that the participants read the explanations, we queried their awareness of the five stylized facts in the survey. The five facts are presented in [Supplementary Material](#).

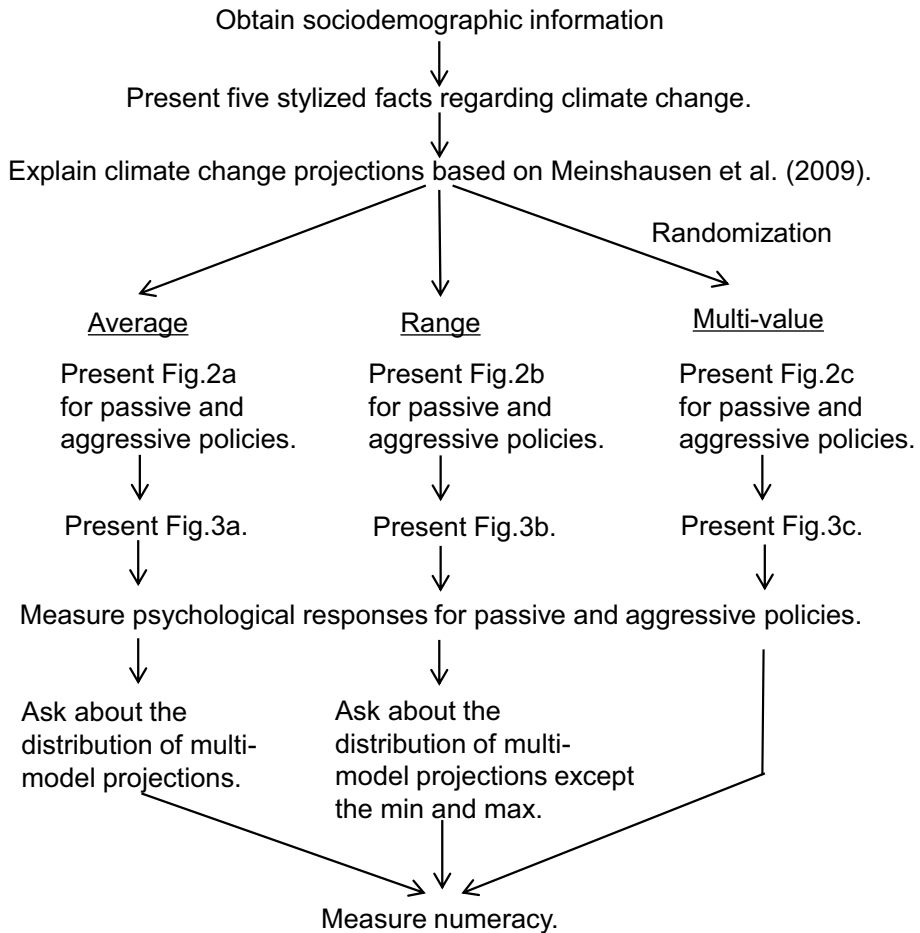
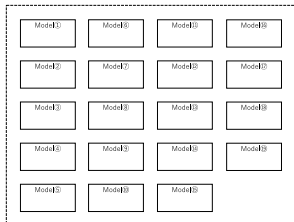


Fig. 1 Outline of the survey process

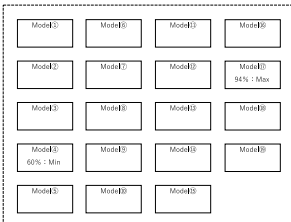
In each format, the multi-model projections for both aggressive and passive policies were explained to the corresponding group using Fig. 2. The average format presented only the average of the projected probabilities and the 19 boxes, corresponding to the 19 models (Fig. 2a). In the range format, two of the 19 boxes contained numbers, corresponding to the model that generated the minimum and maximum probabilities for each policy (Fig. 2b). The multi-value format presented all 19 projected probabilities in each of the 19 boxes for each policy (Fig. 2c). We used redundant illustrations in the range and average formats (i.e., we showed all 19 boxes, but presented data in none, or in two boxes, respectively) to emphasize the existence of other climate models and to control for the visual effect of the illustrations among the three formats. When we consider the three formats simultaneously and compare the range and average formats with the multi-value format in Fig. 2, one may at first notice that the boxes in Fig. 2a and b are blank or incomplete. However, we believe that each participant with the average or range formats did not perceive that the boxes were blank (or incomplete) because they were presented with only one format. Usually, they perceived them as figures explaining the existence of

a. Average format

According to a paper in *Nature*, a worldwide leading scientific journal, the probabilities that the rise of the global surface average temperature will exceed 2 °C in 2100 may be projected with 19 different climate models, if the **policy to maintain current emissions levels** is implemented. **The average probability is 77%** over those 19 projected probabilities. These multiple probabilities are projected because the best available science cannot provide a unique, precise projection.

**b. Range format**

According to a paper in *Nature*, a worldwide leading scientific journal, the probabilities that the rise of the global surface average temperature will exceed 2 °C in 2100 may be projected with 19 different climate models, if the **policy to maintain current emissions levels** is implemented. **The minimum is 60% and the maximum is 94%** among those 19 projected probabilities. These multiple probabilities are projected because the best available science cannot provide a unique, precise projection.

**c. Multi-value format**

According to a paper in *Nature*, a worldwide leading scientific journal, the probabilities that the rise of the global surface average temperature will exceed 2 °C in 2100 may be projected with 19 different climate models, if the **policy to maintain current emissions levels** is implemented. The figure below shows the projected probabilities from the 19 models. These multiple probabilities are projected because the best available science cannot provide a unique, precise projection.

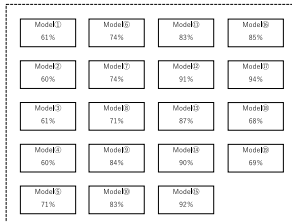


Fig. 2 Explanations of multi-model climate projections using average, range, and multi-value formats The multi-model climate projections of the probabilities of exceeding a 2 °C global temperature increase by the year 2100, given a passive climate policy were explained using **a** average format, **b** range format, and **c** multi-value format. The aggressive policy was explained in the same manner. The participants were randomly assigned to one of the three formats.

19 models.³ Then, the multi-model projections for each policy were concisely presented to the participants as in Fig. 3a, b and c for the average, range, and multi-value formats, respectively.

2.3 Psychological measures

After explaining the multi-model projections for each policy, participants were asked to assess six psychological measures: trust, perceived accuracy, perceived likelihood for the aggressive policy, perceived likelihood for the passive policy, concern for the aggressive policy, and concern for the passive policy. Their numeracy was also assessed using a three-question scale developed by Schwartz et al. (1997).

Trust of the climate model projections was measured by the question “How much would you trust the projected probabilities of exceeding 2 °C in 2100?” A 5-point Likert scale ranging from “Not at all” to “Completely” was used to collect responses.

Perceived accuracy of the climate model projections was measured by the question “How accurate do the projected probabilities of exceeding 2 °C in 2100 seem to you?” A 5-point Likert scale ranging from “Not at all accurate” to “Very accurate” was used to collect responses.

Perceived likelihood for each policy was measured by the question “If the [policy to maintain the current emissions level OR policy to reduce emission level by 50%] was implemented, how likely do you believe it would be that the rise in the global average surface temperature would exceed 2 °C in 2100?” A 7-point Likert scale ranging from “Almost never happen” to “Almost certain” was used to collect responses.

Concern for each policy was assessed using the question “If the [policy to maintain the current emissions level OR policy to reduce the emissions level by 50%] was implemented, to what extent would you be concerned about the climate situation in 2100?” A 5-point

³ We confirmed this by interviewing participants of the pilot surveys—students of Kumamoto University—during the process of designing the survey.

a. Average format

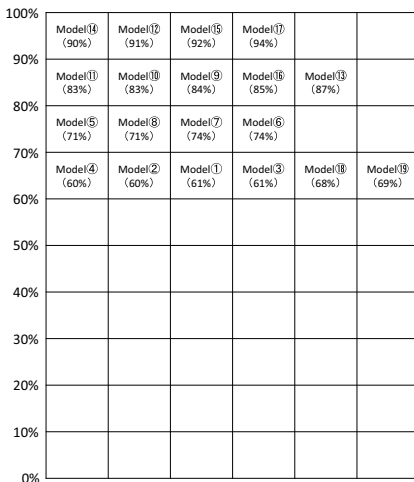
	Policy to maintain GHG current emissions level	Policy to reduce GHG emissions level by 50%
The average probability that the temperature in 2100 exceeds 2 °C	77%	41%

b. Range format

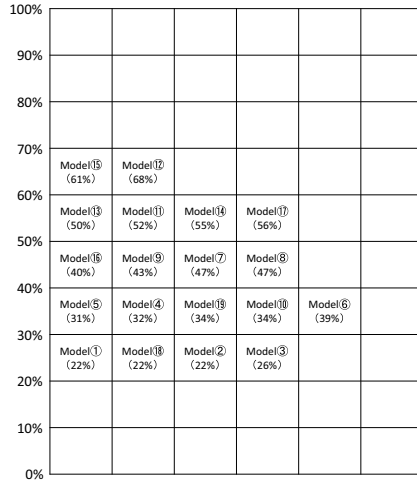
	Policy to maintain GHG current emissions level	Policy to reduce GHG emissions level by 50%
The range of probabilities that the temperature in 2100 exceeds 2 °C	60% – 94%	22% – 68%

c. Multi-value format

The projected probabilities of exceeding 2 °C in 2100 if the policy to maintain current emissions level is implemented.



The projected probabilities of exceeding 2 °C in 2100 if the policy to reduce emissions level by 50% is implemented.



	Policy to maintain GHG current emissions level	Policy to reduce GHG emissions level by 50%
The probabilities that the temperature in 2100 exceeds 2 °C	See left figure above	See right figure above

Fig. 3 Explanation of policy outcomes using three formats. “Policy to maintain GHG current emissions level” is the passive policy and “Policy to reduce GHG emissions level by 50%” is the aggressive policy

Likert scale ranging from “Not at all concerned” to “Very concerned” was used to collected responses.

Numeracy was measured using the three-question scale developed by Schwartz et al. (1997). The participants were asked the following three questions: (1) “Imagine that we flip a fair coin 1,000 times. What is your best guess of how many times the coin would come up heads in 1,000 flips?” (2) “In the BIG BUCKS LOTTERY, the chance of winning a 1,000 Yen prize is 1%. What is your best guess of how many people would win a 1,000 Yen prize if 1,000 people each bought a single ticket to BIG BUCKS?” and (3) “In ACE PUBLISHING SWEEPSTAKES, the chance of winning a car is 1 in 1,000. What percent

of tickets to ACE PUBLISHING SWEEPSTAKES win a car?” The number of correct responses was used as the measure of numeracy (Cronbach’s $\alpha = 0.67$ for the current data).

2.4 Projections other than those provided by the average or range formats

Participants who acknowledged the existence of multi-model projections in the survey may have considered a distribution of the projections other than the ones presented to them, which in turn might have affected their psychological responses. Previous studies showed that individuals considered various distributions of multi-model projections when the range format was used (Dieckmann et al. 2015, 2017; Rinne and Mazzocco 2013). However, these studies did not examine the possibility that they might not have considered any projections other than ones presented to them. To our knowledge, no study examined it when the average format was used after participants acknowledged the existence of multi-model projections.

The survey asked participants who were presented with the average format “How do you think the 19 climate models project the probabilities of exceeding 2 °C in 2100 if the policy to maintain current emissions levels (or the policy to reduce emissions by 50%) is implemented?” They chose one out of six options: (a) many models project equally likely probabilities from 0 to 1, (b) many models project smaller probabilities than the average, (c) many models project larger probabilities than the average, (d) many models project probabilities closer to the average, (e) many models project probabilities quite different from (a) to (d) above, and (f) I did not consider other models. Similarly, we asked participants who were presented with the range format “How do you think the other 17 climate models project the probabilities of exceeding 2 °C in 2100 if the policy to maintain current emissions level (or the policy to reduce emissions by 50%) is implemented?” They chose one out of six options, similar to (a)–(f) above. No further questions were asked of participants who viewed the multi-value format, because these participants were presented with the exact distribution of all modeled probabilities.

2.5 Statistical methods

To examine **RQ1**, we conducted a one-way analysis of variance (ANOVA), in which each of trust, perceived accuracy, perceived likelihood, and concern for each policy were dependent variables and format was the independent variable. Pairwise tests of significant differences used Tukey’s adjustment for multiple comparisons. We explored **RQ2** using an ordinary least squares (OLS) regression model, in which each psychological measure (i.e., trust, perceived accuracy, perceived likelihood, and concern) was regressed on dummy variables for range and multi-value formats (a dummy variable for the average format was omitted, as this format was used as the baseline), interaction terms of format with numeracy and education, and sociodemographic variables. We performed a separate ANOVA and OLS for each dependent variable. All independent continuous variables were centralized to mitigate multicollinearity caused by the interaction terms. Furthermore, we assessed logistic regression models of how numeracy and education affected participants’ consideration of multi-model projections for those other than the ones provided to them. This study considered $p \leq 0.05$ as statistically significant.

Table 1 Descriptive statistics of the survey sample ($N=800$ for each of the three format)

Variable	Definition	Format		
		Average	Range	Multi-value
Numeracy	The number of correct answers to 3 questions on probabilistic concepts	1.78 (1.13)	1.75 (1.14)	1.75 (1.12)
Education	Participant graduated university or above; yes = 1, no = 0	0.47	0.48	0.47
Age	Age of respondent (years)	44.1 (14.1)	44.1 (13.7)	44.8 (13.2)
Female	Female = 1, otherwise = 0	0.49	0.49	0.50
Income	Income (million yen/year)	6.18 (5.22)	5.47 (4.83)	5.47 (4.59)

The table shows sample means. Parentheses indicate standard deviations

3 Results

First, we assessed the random allocation of participants to the different conditions. Table 1 shows that there were only minor differences in the sample means of numeracy and sociodemographic variables among the groups assigned to average, range, and multi-value formats. Table 2 presents the sample mean and standard deviation of each psychological measure. ANOVAs were used to determine whether each psychological measure varied significantly among formats. Post hoc pairwise tests revealed that trust and perceived accuracy with the multi-value format were significantly larger than with the other two formats, although the effect sizes were small (Cohen's d s = 0.119 – 0.201). In addition, the multi-value format was associated with significantly increased perceived likelihood ratings for the passive policy compared to the average format and perceived likelihood and concern for both policies compared to the range format. These effect sizes were also small (Cohen's d s = 0.117 – 0.174). There were no statistically significant differences for any psychological measures between the average and range formats.

Table 3 provides the OLS results for trust and perceived accuracy, with and without interactions between the dummy variables for the range and multi-value formats and each of numeracy and education. Without the interaction terms, the coefficients for multi-value were significant, with values of 0.171 for trust and 0.145 for perceived accuracy. This confirms that the multi-value format increased trust and perceived accuracy compared to the average format, even after adjusting for numeracy, education, and sociodemographic variables. Coefficients for trust and perceived accuracy were not significant for the range format. With the interaction terms, the coefficient for Multi-value $\times$ Education was 0.193 for trust, which was statistically significant, and the coefficients for Range $\times$ Education and Multi-value $\times$ Education were 0.148 and 0.162, respectively, which approached significance. These results suggest that education moderated the effect of format. The other interaction terms were not statistically significant. Figure 4 shows that trust and perceived accuracy of the more-educated participants who were presented with the multi-value format were clearly larger than those of the other formats, while the multi-value format was associated with slightly increased trust and perceived accuracy among less-educated participants relative to the average and range formats.

Table 4 presents the OLS results for perceived likelihood and concern for passive and aggressive policies with and without the interaction terms. The latter results show

Table 2 Results of ANOVAs and post hoc pairwise tests

Variable	ANOVA			Post hoc pairwise test	
	Average <i>n</i> = 800	Range <i>n</i> = 800	Multi-value <i>n</i> = 800	Average vs. Range	Average vs. Multi-value
Trust	2.33 (0.84)	2.35 (0.91)	2.50 (0.88)	$p = 0.878$ $d = 0.024$	$p < 0.001$ $d = 0.201$
Perceived accuracy	2.34 (0.85)	2.38 (0.91)	2.49 (0.87)	$p = 0.596$ $d = 0.048$	$p = 0.002$ $d = 0.173$
Perceived likelihood for passive policy	3.55 (1.31)	3.51 (1.46)	3.76 (1.34)	$p = 0.875$ $d = -0.024$	$p = 0.006$ $d = 0.159$
Perceived likelihood for aggressive policy	3.30 (1.26)	3.26 (1.39)	3.42 (1.27)	$p = 0.824$ $d = -0.029$	$p = 0.170$ $d = 0.093$
Concern for passive policy	2.56 (1.01)	2.48 (1.06)	2.64 (1.02)	$p = 0.301$ $d = -0.074$	$p = 0.267$ $d = 0.079$
Concern for aggressive policy	2.41 (0.93)	2.35 (1.00)	2.48 (0.94)	$p = 0.364$ $d = -0.067$	$p = 0.408$ $d = 0.065$

Parentheses indicate standard deviations. Trust, perceived accuracy, perceived likelihood, and concern for each policy were dependent variables; format was the independent variable. Pairwise tests were conducted with Tukey's adjustment for multiple comparisons. *p* values < 0.05 are emphasized in bold. All *d* values are Cohen's *d*

Table 3 Results of OLS regression for trust and perceived accuracy

Variables	Trust		Perceived accuracy	
Range	0.024 (0.043)	− 0.056 (0.061)	0.043 (0.043)	− 0.021 (0.061)
Range × Numeracy		0.025 (0.039)		− 0.010 (0.039)
Range × Education		0.137 (0.088)		0.148 + (0.088)
Multi-value	0.171** (0.043)	0.080 (0.060)	0.145** (0.043)	0.070 (0.060)
Multi-value × Numeracy		0.001 (0.039)		− 0.005 (0.039)
Multi-value × Education		0.193* (0.088)		0.162 + (0.088)
Numeracy	0.093** (0.016)	0.085** (0.028)	0.114** (0.016)	0.119* (0.028)
Education	0.059 (0.037)	− 0.051 (0.063)	0.058 (0.037)	− 0.045 (0.063)
Age	0.002 (0.001)	0.002 (0.001)	0.002 + (0.001)	0.002 + (0.001)
Female	0.249** (0.037)	0.248** (0.037)	0.224** (0.037)	0.223** (0.037)
Income	0.001 (0.004)	0.001 (0.004)	− 0.003 (0.004)	− 0.003 (0.004)
Constant	2.176** (0.041)	2.183** (0.049)	2.135** (0.042)	2.182** (0.049)
<i>N</i>	2400	2400	2400	2400
<i>R</i> ²	0.038	0.040	0.038	0.040
Adjusted <i>R</i> ²	0.035	0.036	0.035	0.035

Parentheses indicate standard errors. Range (multi-value) is a dummy variable that was 1 if a participant was presented with the range (multi-value) format, 0 otherwise. A dummy variable for the average format was omitted, as this format was used as the baseline. All independent continuous variables were centralized to mitigate multicollinearity. The VIF of each independent variable was < 3.4

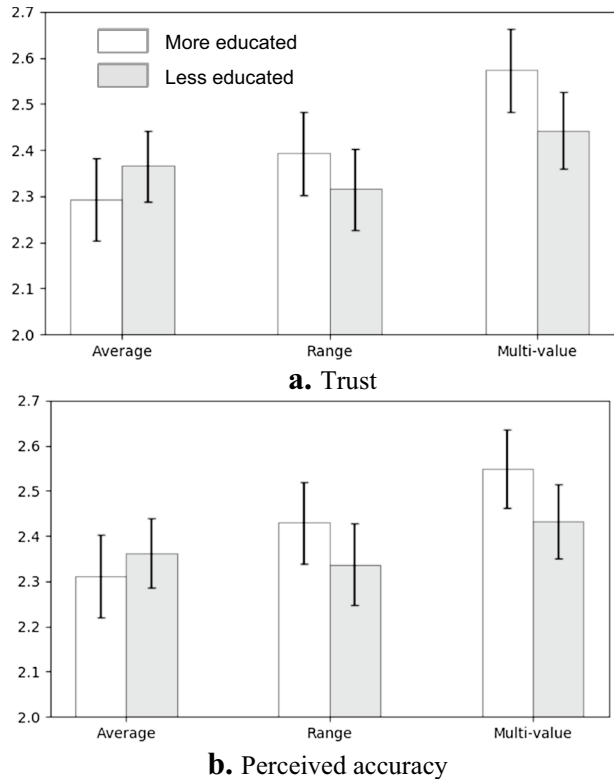
** $p < 0.01$, * $p < 0.05$, + $p < 0.10$

that the coefficients for perceived likelihood for the multi-value format were 0.204 for the passive policy after adjusting for numeracy, education, and sociodemographic variables; the former value was statistically significant. Neither range nor multi-value was significantly associated with concern for both policies. No interaction terms were statistically significant for both perceived likelihood and concern.

Furthermore, we examined how participants assessed the distribution of multi-model projections when they were presented with the average and range formats. Although we explained the existence of 19 climate model projections clearly and repeatedly in the survey, Table 5 indicates that approximately 55% of the participants never considered multi-model projections other than those provided in the average and range formats. The other 45% of the participants considered various distributions of multi-model projections, consistent with previous studies (Dieckmann et al. 2015, 2017; Rinne and Mazzocco 2013).

We performed a logistic regression to explore the relationship between numeracy or education and participants' consideration of multi-model projections other than those presented in the average or range formats. Each dependent variable was a dummy variable that was 1 if participants chose any option from (a) to (e) and 0 if they chose (f) "I did not consider other models" shown in Table 5. Independent variables were numeracy, education, and the other sociodemographic variables as covariates (Table 1). The results indicated that numeracy was not significantly associated with consideration of other model projections for all cases. In contrast, the odds ratios for education for the average format were 1.399 for the passive policy ($SE = 0.212$, $p = 0.027$; $n = 800$, $Pseudo R^2 = 0.018$) and 1.343 for the aggressive policy ($SE = 0.204$, $p = 0.052$; $n = 800$,

Fig. 4 Trust and perceived accuracy for the three formats for more- and less-educated participants. Bars indicate means and whiskers represent 95% CIs



*Pseudo R*²=0.020), while for the range format, these values were 1.816 (*SE*=0.281, *p*<0.001; *n*=800, *Pseudo R*²=0.030) and 1.785 (*SE*=0.275, *p*<0.001; *n*=800, *Pseudo R*²=0.025), respectively. This suggests that more-educated people considered multi-model projections other than those presented in the average and range formats to a greater extent than did less-educated participants.

4 Discussion

Climate change projections are generally obtained from ensemble methods, which provide uncertainty in that different models typically produce different projections. Such uncertainty should be communicated to the general public in the interests of transparency and honesty. This study explored the average, range, and multi-value formats for communicating uncertainty. First, we investigated how the average, range, and multi-value formats affected trust, perceived accuracy, perceived likelihood, and concern regarding multi-model climate projections (RQ1). The results indicated that the multi-value format increased trust and perceived accuracy of climate projections compared to the average and range formats. After acknowledging the uncertainty of climate projections, the participants might have considered the multi-value format, which provided all projections, as more honest and informative than the average and range formats, which provided only one or

Table 4 Results of OLS regression for perceived likelihood and concern

Variables	Perceived likelihood		Concern	
	Passive	Aggressive	Passive	Aggressive
Range	−0.034 (0.068)	−0.130 (0.096)	−0.1077 (0.093)	−0.118 + (0.070)
Range × Numeracy	0.081 (0.061)	−0.042 (0.065)	0.071 (0.059)	0.063 (0.045)
Range × Education	0.103 (0.138)		0.051 (0.133)	0.085 (0.103)
Multi-value	0.204** (0.068)	0.118 (0.096)	0.034 (0.090)	0.022 (0.070)
Multi-value × Numeracy	−0.010 (0.061)	0.111 + (0.065)	0.001 (0.059)	0.013 (0.046)
Multi-value × Education	0.194 (0.138)		0.161 (0.133)	0.109 (0.103)
Numeracy	0.151** (0.025)	0.127** (0.044)	−0.006 (0.042)	0.131** (0.019)
Education	0.046 (0.059)	−0.053 (0.099)	−0.027 (0.095)	0.106** (0.032)
Age	0.003 (0.002)	0.003 (0.002)	0.001 (0.002)	0.003 (0.074)
Female	0.338** (0.058)	0.340** (0.058)	0.001 (0.002)	0.006** (0.002)
Income	−0.005 (0.006)	−0.006 (0.006)	0.271** (0.055)	0.003 (0.001)
Constant	3.271** (0.066)	3.331** (0.077)	0.355** (0.043)	0.291** (0.040)
<i>N</i>	2400	2400	−0.001 (0.004)	0.001 (0.004)
<i>R</i> ²	0.032	0.034	2.276** (0.049)	2.192** (0.046)
Adjusted <i>R</i> ²	0.029	0.030	2400	2.204** (0.054)
			0.014	2400
			0.051	0.028
			0.048	0.025

Parentheses indicate standard errors. Range (multi-value) is a dummy variable that was 1 if a participant was presented with the range (multi-value) format or 0 otherwise. A dummy variable for the average format was omitted, as this format was used as the baseline. All independent continuous variables were centralized to mitigate multicollinearity. The VIF of each independent variable was < 3.4

***p* < 0.01, **p* < 0.05, + *p* < 0.10

Table 5 Multi-model climate projections considered other than those provided in the average and range formats
The options for the average format (for the range format)

	Average		Range	
	Passive	Aggressive	Passive	Aggressive
(a) Many models project equally likely probabilities from 0 to 1 (equally likely within the range)	0.08	0.06	0.09	0.06
(b) Many models project smaller probabilities than the average (closer to the minimum)	0.08	0.06	0.07	0.05
(c) Many models project larger probabilities than the average (closer to the maximum)	0.10	0.14	0.09	0.14
(d) Many models project probabilities closer to the average (to the middle)	0.12	0.14	0.12	0.14
(e) Many models project probabilities quite different from (a) to (d) above	0.07	0.06	0.06	0.05
(f) I did not consider other models	0.56	0.56	0.57	0.55
<i>N</i>	800	800	800	800

The table indicates the proportion of participants choosing each option among the six options that describe how the multi-model projections are distributed for passive and aggressive policies

two projections among the complete set. This result is consistent with Howe et al. (2019), who found that public trust might be enhanced by communicating more climate projections. They found that the range format, which presented the worst and best projections, increased public trust in climate scientists, compared to the half-range format, which conveyed only the worst projection.

The results indicated that the multi-value format also increased perceived likelihood and concern for both policies compared to the range format, and perceived likelihood for the passive policy compared to the average format. This suggests that communicating all climate model projections using the multi-value format might enhance perceived likelihood and concern regarding climate projections compared to communicating the average projection or the best and worst projections with the range format. It is desirable to enhance public support for climate policies; communicating all projections might help the general public's perception of the likelihood of climate change and their concern regarding climate change better correspond to scientific projections. However, there exists the alternative possibility that communicating all projections might result in overestimated likelihood and excessive concern due to the difficulty of understanding the multi-value format. The former possibility is supported by Bosetti et al.'s (2017) findings that presenting all climate models' projections led the participants' subjective probability estimates of the increase in global temperature by the year 2100 to be closer to those suggested by climate science.

Table 2 shows that the effect sizes of multi-value format were small. This might explain the fact that the participants' psychological responses were determined by various factors such as their abilities, beliefs, and attitudes and that the differences in the formats were part of them. The effects of multi-value format were clearer in participants' trust and perceived accuracy compared to their perceived likelihood and concern. This might be due to the following reason. Trust and perceived accuracy were affected largely by how to communicate climate projections, while perceived likelihood and concern were influenced mainly by the outcomes of climate projections. The format effects might be closely associated with the mode of communication rather than the outcomes of climate projections, which might result in clearer format effects on trust and perceived accuracy than on perceived likelihood and concern.

The current results revealed no differences between average and range formats for all psychological responses. This appears inconsistent with previous research (Han et al. 2011; Howe et al. 2019; Joslyn and LeClerc 2016). The reason for the difference might be that these previous studies communicated the average estimate with certainty, whereas the current study conveyed the average estimate while acknowledging the uncertainty of the projections. This suggests that the average and range formats might engender minimal differences in ratings of trust, perceived credibility, perceived likelihood, and concern regarding climate projections, after acknowledging the uncertainty of such projections.

Second, this study explored how numeracy and education moderated the effect of each format on general public trust, perceived accuracy, perceived likelihood, and concern regarding climate projections (RQ2). Contrary to our expectation, the results implied that numeracy did not moderate the effect of format on all psychological responses. This is consistent with Han et al. (2011), who found that numeracy did not moderate the effect of multi-model projections on risk perception and worry for cancer risks. In contrast, education increased the effect of the multi-value format on enhancing trust and perceived accuracy, and the effect of the range format on enhancing perceived accuracy. This is possibly because the effects of the multi-value format might be associated with contextual understanding to comprehend that the multi-value format honestly and transparently provided all

projections obtained from the multiple climate models. Contextual understanding may be related to the education level rather than numeracy.

Figure 4 indicates that even less-educated participants perceived slightly greater trust and accuracy from multi-model projections compared to average and range formats, while such effects were clearly among more-educated participants. These results are consistent with Joslyn and LeClerc (2016), who found that the range format enhanced trust among more-educated people but did not significantly affect less-educated people. Table 5 indicates that more than half of our participants did not consider other projections. The results of the logistic regression imply that better-educated participants were more likely than less-educated participants to consider climate projections other than those provided via the average and range formats. Thus, more-educated people might express more trust and perceive greater accuracy from the multi-value format than the other two formats because such people are more likely than less-educated people to notice climate projections that are not communicated via the average and range formats. Regarding perceived likelihood and concern, education did not significantly influence the format effects, consistent with Joslyn and LeClerc (2016), who also found that the format did not differentially affect the concern responses of more- and less-educated people.

In summary, the current study suggests that the different formats by which climate projections are presented might affect the general public's trust, perceived accuracy, perceived likelihood, and concern, although the effect sizes are small. Specifically, the multi-value format might increase trust and perceived accuracy of climate projections compared to the average and range formats. The effect of the multi-value format appears to not be influenced by numeracy. The multi-value format might increase trust and perceived accuracy among more-educated people compared to the other two formats, while less-educated people might respond similarly to the three formats. These findings have the policy implication that the multi-value format might be better than the average and range formats for enhancing trust and perceived accuracy, regardless of the numeracy and education level of the recipients of the information. In addition, the multi-value format might also increase the general public's perceived likelihood and concern regarding climate projections compared to the range format, and perceived likelihood compared to the average format, regardless of numeracy and education level. Combined with previous findings that individuals' trust, perceived accuracy, perceived likelihood, and concern regarding climate projections may increase public support for climate change mitigation policies (e.g., Drews and van den Bergh 2015; van der Linden 2017), our results suggest that the multi-value format might enhance public support compared to the average and range formats for communicating the uncertainty inherent in ensemble methods. In return for such advantages, the multi-value format may generally require a larger space to present and explain multiple projections in the text or in the media. Thus, in practice, the multi-value format should be used only for communicating uncertainty about the projection of the event that is critically relevant to the climate policy goal such as rise in the global average surface temperature.

There are several limitations to this study as well as scope for further research. First, due to the exploratory nature of this study, we had to make slightly arbitrary choices for formats, visual presentations, and explanations in the survey. Thus, we focused on the most common format (range format) and two extreme ones (average and multi-value formats). The results suggested that the average or range formats provide insufficient information for enhancing trust and perceived accuracy compared to the multi-value format. However, this does not necessarily imply that the multi-value format is the best format. Some simpler formats can have a similar effect as the multi-value format. For example, presenting both average and range estimates or using a boxplot format are promising approaches that need to

be investigated. Second, our results were obtained from Japanese participants alone. Previous studies have reported that Japanese people have high levels of awareness, concern, and attitude toward addressing climate change, similar to other developed countries in North America and Europe (Brechin and Bhandari 2011; Lee et al. 2015). This suggests the possibility of extending our findings to people from such countries. Future studies should explore the generalizability of our findings across cultures. Third, we could not identify a psychological mechanism explaining why the multi-value format increased individuals' trust, perceived accuracy, perceived likelihood, and concern regarding the climate projections, although we discussed some possible reasons for the increases. Fourth, this study included only numeracy and education as potential moderators of the format effects. Other factors should be considered in the future, including world views and political attitudes. Thus, further research is required to investigate why the multi-value format increased the perceived credibility of climate change estimates and perceptions of the risk of climate change.

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Data availability All data and materials as well as STATA codes for statistical analysis are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

References

- Benjamin DM, Budescu DN (2018) The role of type and source of uncertainty on the processing of climate models projections. *Front Psychol* 9:403. <https://doi.org/10.3389/fpsyg.2018.00403>
- Bosetti V, Weber E, Berger L, Budescu D, Liu N, Tavoni M (2017) COP21 climate negotiators' responses to climate model forecasts. *Nat Clim Chang* 7:185–189. <https://doi.org/10.1038/nclimate3208>
- Brechin SR, Bhandari M (2011) Perceptions of climate change worldwide. *WIREs Clim Change* 2:871–885. <https://doi.org/10.1002/wcc.146>
- Budescu DV, Broomell SB, Lempert RJ, Keller K (2014) Aided and unaided decisions with imprecise probabilities in the domain of losses. *EURO J Decis Processes* 2:31–62. <https://doi.org/10.1007/s40070-013-0023-4>
- Ceci SJ (1991) How much does schooling influence general intelligence and its cognitive components? A reassessment of the evidence. *Dev Psychol* 27:703–722. <https://doi.org/10.1037/0012-1649.27.5.703>
- Drewns S, van den Bergh JCJM (2015) What explains public support for climate policies? A review of empirical and experimental studies. *Clim Policy* 16:855–876. <https://doi.org/10.1080/14693062.2015.1058240>
- Dieckmann NF, Slovic P, Peters E (2009) The use of narrative evidence and explicit probability by decision makers varying in numeracy. *Risk Anal* 29:1473–1488. <https://doi.org/10.1111/j.1539-6924.2009.01279.x>

- Dieckmann NF, Peters E, Gregory R (2015) At home on the range? Lay interpretations of numerical uncertainty ranges. *Risk Anal* 35:1281–1295. <https://doi.org/10.1111/risa.12358>
- Dieckmann NF, Gregory R, Peters E, Hartman R (2017) Seeing what you want to see: How imprecise uncertainty range enhances motivated reasoning. *Risk Anal* 37:471–486. <https://doi.org/10.1111/risa.12639>
- Dietz T, Dan A, Shwom R (2007) Support for climate change policy: Social psychological and social structural influences. *Rural Sociol* 72:185–214. <https://doi.org/10.1526/003601107781170026>
- Fischhoff B, Davis AL (2014) Communicating scientific uncertainty. *Proc Natl Acad Sci* 111:13664–13671. <https://doi.org/10.1073/pnas.1317504111>
- Han PK, Klein WM, Lehman TC, Massett H, Lee SC, Freedman AN (2009) Laypersons' responses to the communication of uncertainty regarding cancer risk estimates. *Med Decis Mak* 29:391–403. <https://doi.org/10.1177/0272989X08327396>
- Han PK, Klein WM, Lehman TC, Massett H, Lee SC, Freedman AN (2011) The communication of uncertainty regarding individualized cancer risk estimates: Effects and influential factors. *Med Decis Mak* 31:354–366. <https://doi.org/10.1177/0272989X10371830>
- Howe LC, MacInnis B, Krosnick JA, Markowitz EM, Socolow R (2019) Acknowledging uncertainty impacts public acceptance of climate scientists' predictions. *Nat Clim Chang* 9:863–867. <https://doi.org/10.1038/s41558-019-0587-5>
- IPCC (2013) Climate change 2013: the physical science basis. In: Stocker TF, Qin D, Plattner G-K, Tignor M, Allen SK, Boschung J, Nauels A, Xia Y, Bex V, Midgley PM (eds) Contribution of working group I to the fifth assessment report of the intergovernmental panel on climate change. Cambridge University Press, Cambridge, pp 138–142. <https://doi.org/10.1017/CBO9781107415324>
- IPCC (2014) Summary for policymakers. In: Edenhofer O, Pichs-Madruga R, Sokona Y, Farahani E, Kadner S, Seyboth K, Adler A, Baum I, Brunner S, Eickemeier P, Kriemann B, Savolainen J, Schlömer S, vonStechow C, Zwickel T, Minx JC (eds) Climate change 2014: mitigation of climate change. contribution of working group III to the fifth assessment report of the intergovernmental panel on climate change. Cambridge University Press, Cambridge, pp 1–30
- Jensen JD (2008) Scientific uncertainty in news coverage of cancer research: Effects of uncertainty on scientists' and journalists' credibility. *Hum Commun Res* 34:347–369. <https://doi.org/10.1111/j.1468-2958.2008.00324.x>
- Johnson BB (2003) Further notes on public response to uncertainty in risks and science. *Risk Anal* 23:781–789. <https://doi.org/10.1111/1539-6924.00355>
- Johnson BB, Slovic P (1995) Presenting uncertainty in health risk assessment: Initial studies of its effects on risk perception and trust. *Risk Anal* 15:485–494. <https://doi.org/10.1111/j.1539-6924.1995.tb00341.x>
- Joslyn S, LeClerc JE (2016) Climate projections and uncertainty communication. *Top Cogn Sci* 8:222–241. <https://doi.org/10.1111/tops.12177>
- Kunreuther H, Heal G, Allen M, Edenhofer O, Field CB, Yohe G (2013) Risk management and climate change. *Nat Clim Chang* 3:447–450. <https://doi.org/10.1038/nclimate1740>
- Lee TM, Markowitz EM, Howe PD, Ko CY, Leiserowitz AA (2015) Predictors of public climate change awareness and risk perception around the world. *Nat Clim Chang* 5:1014–1020. <https://doi.org/10.1038/nclimate2728>
- Meinshausen M, Meinshausen N, Hare W, Raper SC, Frieler K, Knutti R, Frame DJ, Allen MR (2009) Greenhouse-gas emission targets for limiting global warming to 2 °C. *Nature* 458:1158–1162. <https://doi.org/10.1038/nature08017>
- Mossler MV, Bostrom A, Kelly RP, Crosman KM, Moy P (2017) How does framing affect policy support for emissions mitigation? Testing the effects of ocean acidification and other carbon emissions frames. *Glob Environ Chang* 45:63–78. <https://doi.org/10.1016/j.gloenvcha.2017.04.002>
- O'Connor RE, Bord RJ, Fisher A (1999) Risk perceptions, general environmental beliefs, and willingness to address climate change. *Risk Anal* 19:461–471. <https://doi.org/10.1023/A.1007004813446>
- O'Connor RE, Bord RJ, Yarnal B, Wiefek N (2002) Who wants to reduce greenhouse gas emissions? *Soc Sci Q* 83:1–17. <https://doi.org/10.1111/1540-6237.00067>
- Patt A (2007) Assessing model-based and conflict-based uncertainty. *Glob Environ Chang* 17:37–46. <https://doi.org/10.1016/j.gloenvcha.2006.10.002>
- Peters E (2012) Beyond comprehension: the role of numeracy in judgments and decisions. *Curr Dir Psychol Sci* 21:31–35. <https://doi.org/10.1177/0963721411429960>
- Peters E, Hart S, Fraenkel L (2011) Informing patients: the influence of numeracy, framing, and format of side-effect information on risk perceptions. *Med Decis Mak* 31:432–436. <https://doi.org/10.1177/0272989X10391672>
- Peters E, Västfjäll D, Slovic P, Mertz CK, Mazzocco K, Dickert S (2006) Numeracy and decision making. *Psychol Sci* 17:407–413. <https://doi.org/10.1111/j.1467-9280.2006.01720.x>

- Politi MC, Han PKJ, Col NF (2007) Communicating the uncertainty of harms and benefits of medical interventions. *Med Decis Mak* 27:681–695. <https://doi.org/10.1177/0272989X07307270>
- Rinne LF, Mazzocco MMM (2013) Inferring uncertainty from interval estimates: Effects of alpha level and numeracy. *Judgm Decis Mak* 8:330–344
- Ritchie SJ, Tucker-Drob EM (2018) How much does education improve intelligence? A meta-analysis. *Psychol Sci* 29:1358–1369. <https://doi.org/10.1177/0956797618774253>
- Rothman RL, Montori VM, Cherrington A, Pigone MP (2008) Perspective: the role of numeracy in health-care. *J Health Commun* 13:583–595. <https://doi.org/10.1080/10810730802281791>
- Reyna VF, Nelson WL, Han PK, Dieckmann NF (2009) How numeracy influences risk comprehension and medical decision making. *Psychol Bull* 135:943–973. <https://doi.org/10.1037/a0017327>
- Schapira MM, Nattinger AB, McHorney CA (2001) Frequency or probability? A qualitative study of risk communication formats used in health care. *Med Decis Mak* 21:459–467. <https://doi.org/10.1177/0272989X0102100604>
- Schwartz LM, Woloshin S, Black WC, Welch HG (1997) The role of numeracy in understanding the benefit of screening mammography. *Ann Int Med* 1:966–972. <https://doi.org/10.7326/0003-4819-127-11-199712010-00003>
- Sheridan SL, Pignone M (2002) Numeracy and the medical student's ability to interpret data. *Eff Clin Pract* 5:35–40. <https://doi.org/10.1037/a0017327>
- Sheridan SL, Pignone MP, Lewis CL (2003) A randomized comparison of patients' understanding of number needed to treat and other common risk reduction formats. *J Gen Int Med* 18:884–892. <https://doi.org/10.1046/j.1525-1497.2003.21102.x>
- Smith N, Leiserowitz A (2013) American evangelicals and global warming. *Glob Environ Chang* 23:1009–1017. <https://doi.org/10.1016/j.gloenvcha.2013.04.001>
- Spence A, Poortinga W, Butler C, Pidgeon NF (2011) Perceptions of climate change and willingness to save energy related to flood experience. *Nat Clim Chang* 1:46–49. <https://doi.org/10.1038/nclimate1059>
- Spiegelhalter D (2017) Risk and uncertainty communication. *Annu Rev Stat Appl* 4:31–60. <https://doi.org/10.1146/annurev-statistics-010814-020148>
- Spiegelhalter D, Pearson M, Short I (2011) Visualizing uncertainty about the future. *Science* 333:1393–1400. <https://doi.org/10.1126/science.1191181>
- Stern N (2007) *The economics of climate change: the Stern review*. Cambridge University Press, Cambridge
- Tebaldi C, Knutti R (2007) The use of the multi-model ensemble in probabilistic climate projections. *Philos Trans R Soc* 365:2053–2075. <https://doi.org/10.1098/rsta.2007.2076>
- Trevena LJ, Zikmund-Fisher BJ, Edwards A, Gaissmaier W, Galesic M, Han PK, King JL, Lawson ML, Linder SK, Lipkus I, Ozanne E, Peters E, Timmermans D, Woloshin S (2013) Presenting quantitative information about decision outcomes: a risk communication primer for participant decision aid developers. *BMC Med Inform Decis Mak* 13(Suppl 2):S7. <https://doi.org/10.1186/1472-6947-13-S2-S7>
- van der Linden S (2017) Determinants and measurement of climate change risk perception, worry, and concern. *Oxford Research Encyclopedia of Climate Science*. <https://doi.org/10.1093/acrefore/9780190228620.013.318>
- Viscusi WK (1997) Alarmist decisions with divergent risk information. *Econ J* 107:1657–1670. <https://doi.org/10.1111/j.1468-0297.1997.tb00073.x>
- Zahran S, Brody SD, Grover H, Vedlitz A (2006) Climate change vulnerability and policy support. *Soc Nat Resour* 19:771–789. <https://doi.org/10.1080/08941920600835528>

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